Machine-learning emulation of precipitation $\delta^{18}O$ reveals its cross-model transferability across PMIP4 palaeoclimate simulations

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Abstract

Precipitation $\delta^{18}O$ links climate models to a wide range of palaeoclimate proxies, yet most models in the Palaeoclimate Modelling Intercomparison Project do not carry water isotopes and cannot be compared with isotope records directly. An emulator of precipitation $\delta^{18}O$ is trained here on the 8199-year isotope-enabled AWIESM2-wiso transient simulation, with seven regression methods from four learner families relating monthly $\delta^{18}O$ to fourteen standard climate fields. On years withheld from training the emulator recovers the interannual variability of the precipitation-weighted annual $\delta^{18}O$ at a land-mean temporal correlation of 0.75. The correlation remains at 0.66 when the emulator is transferred to MPI-ESM-wiso, an independent isotope-enabled model, and the mid-Holocene minus pre-industrial change of that model is reproduced at a spatial correlation of 0.87. Internal fit turns out to be a poor guide to transfer, because kernel and neural learners match the tree ensembles inside the training model and then lose most of their skill on a foreign one. We find that predicting anomalies instead of absolute values restores all seven methods to correlations between 0.85 and 0.92, which identifies the removal of the inter-model base state, rather than the learner family, as the main control on transferability. Applied to nine PMIP4 mid-Holocene simulations the emulator produces a pronounced depletion over the Sahara and the Sahel that at least seven of the nine models share, the isotopic expression of the Green Sahara. Changing the climate model moves the reconstructed change three times as much as changing the learning method. The emulated fields reproduce the spatial temperature effect and its latitude dependence, while the temporal slopes between the two climate states reach only about half of the spatial ones, which cautions against calibrating a palaeothermometer on modern gradients alone.

1 Introduction

The oxygen isotopic composition of precipitation, expressed as $\delta^{18}O$, records the integrated history of evaporation, transport and rain-out along an air mass trajectory (Dansgaard, 1964). Speleothems, ice cores and other archives preserve this signal, which makes precipitation $\delta^{18}O$ a central quantity for comparing climate models against palaeoclimate proxies. A meaningful comparison requires the model to carry water isotopes as prognostic tracers. Only a small number of isotope-enabled general circulation models exist, because the implementation and the added computational cost are considerable (Cauquoin et al., 2019; Shi et al., 2023). Most simulations contributed to the Palaeoclimate Modelling Intercomparison Project therefore output temperature, precipitation and circulation, but not $\delta^{18}O$, and cannot be placed alongside isotope records without an intermediate step.

Two routes have been used to bridge this gap. Isotope-enabled models compute $\delta^{18}O$ explicitly and allow direct proxy comparison, at the price of running a dedicated model version (Cauquoin et al., 2019; Shi et al., 2023; Bühler et al., 2021). Proxy system models and data assimilation instead translate between climate fields and proxy space through forward operators or statistical priors (Dee et al., 2015; Hakim et al., 2016). A learned emulator offers a third route. If the mapping from climate variables to $\delta^{18}O$ can be learned from a model that does carry isotopes, the same mapping can then be applied to the climate fields of a model that does not. Learned emulators of standard climate fields have become a mature tool that reproduces model output at a small fraction of the cost (Tebaldi et al., 2025; Cresswell-Clay et al., 2025), yet water isotopes have remained outside their scope.

Machine learning has instead been applied to water isotopes mostly as a spatial interpolation problem. Random-forest and related methods predict station or gridded precipitation $\delta^{18}O$ from meteorological and geographic predictors across Europe (Nelson et al., 2021; Erdélyi et al., 2023), Australia (Falster et al., 2026), China (Cui et al., 2025; Wang et al., 2025; Liu et al., 2025) and the Arctic Ocean (Klein et al., 2024), which produces useful isoscapes but does not emulate a climate model. The one study that emulated the isotope field of a general circulation model used a convolutional network within a single last-millennium simulation (Wider et al., 2023). It reported that the network skill fell to the level of pixel-wise linear regression when the emulator was applied to a different climate model, which leaves the cross-model problem open. Whether an emulator trained on one isotope-enabled model can be transferred to others, and how much skill survives that transfer, has not been quantified.

In the present study we train an emulator of precipitation $\delta^{18}O$ on a long isotope-enabled transient simulation and test its transfer to independent climate models. Seven regression methods drawn from four learner families are compared, the emulator is validated against a second isotope-enabled model that supplies ground truth, and it is then applied to nine PMIP4 mid-Holocene simulations that lack isotopes. The transfer is quantified against a self-reconstruction upper bound. The uncertainty of the emulated product is separated into a contribution from the choice of climate model and a contribution from the choice of learning method, and the emulated fields are finally examined for the classical isotope-climate relations they should contain if the learned mapping is physical.

2 Results

2.1 Skill within the training model

The upper bound of what the emulator can achieve is set by applying it back to the simulation it was trained on. One model year in eight enters the training set, which leaves 7174 of the 8199 transient years unseen, and the skill is evaluated on those years alone. All seven learners recover the interannual variability of the precipitation-weighted annual $\delta^{18}O$ over most of the land surface (Fig. 1). The land-mean temporal correlation reaches 0.75 for gradient boosting and histogram gradient boosting, 0.74 for the random forest and the neural network, 0.71 for the support vector regression, 0.69 for kernel ridge and 0.64 for the linear baseline. Skill is highest over the monsoon domains and the mid-latitude continents and falls in the dry subtropical interiors, where the precipitation weighting rests on few wet months.

The same behaviour holds for the low-frequency evolution of the transient. Regional precipitation-weighted $\delta^{18}O$ follows the orbital-scale trends of the eight thousand years closely, and the seven methods stay within a narrow band around the truth, with regional correlations between 0.46 and 0.97 and biases within 0.15 ‰ (Fig. S1). This self-reconstruction carries no inter-model offset, so it isolates the part of the error that comes from the learning problem itself. Everything that follows is measured against it.

2.2 Transfer to an independent isotope-enabled model

MPI-ESM-wiso provides the independent test, since it simulates its own precipitation δ^{18}O and never enters the training. The emulator receives only its climate fields and returns δ^{18}O, which is then compared with the isotopes the model itself computed. Over the mid-Holocene the land-mean temporal correlation is 0.66 for gradient boosting, 0.65 for the random forest and 0.64 for histogram gradient boosting (Fig. 2). Repeating the exercise on the pre-industrial state of the same model gives 0.64, 0.63 and 0.62, so the transfer does not depend on the base state to which the emulator is applied. The zonal profiles show that the loss is concentrated in the tropics and the subtropics, while the extratropical continents retain correlations near 0.7.

Against the self-reconstruction bound of 0.75 the cross-model value of 0.66 puts the structural penalty of learning δ^{18}O in one model and predicting it in another at about 0.1 in correlation. The penalty has a systematic component that the correlation alone does not reveal. Regional time series of the raw reconstruction sit parallel to the truth but offset from it, by up to 12.2 ‰ for the linear baseline over Africa (Fig. S2). The offset is a difference in base state between the two models rather than a failure to track variability, and this distinction motivates the anomaly framing examined next.

2.3 Learning method and preprocessing framing

Internal fit is a poor guide to cross-model skill. Within the training model the kernel and neural learners fit the δ^{18}O field about as well as the tree ensembles, with validation errors of 1.73 ‰ for the neural network, 1.79 ‰ for the support vector regression and 1.85 ‰ for kernel ridge, against 1.27 ‰ for gradient boosting, 1.69 ‰ for histogram gradient boosting and 1.70 ‰ for the random forest, and all of them ahead of the linear fit at 2.01 ‰. Under transfer the ranking inverts. The precipitation-weighted temporal correlation on MPI-ESM-wiso holds between 0.61 and 0.64 for the three tree methods but falls to 0.32 for the support vector regression, 0.38 for kernel ridge and 0.38 for the neural network, all of them below the linear baseline at 0.45 (Fig. 5). Learners that fit the training climate closely therefore need not carry any of that fit to a foreign model.

The contrast sharpens on the mid-Holocene minus pre-industrial change, the quantity a palaeoclimate application actually needs. In the raw framing the tree methods reproduce the true MPI-ESM-wiso change at correlations between 0.84 and 0.87, whereas kernel ridge collapses to −0.02, the support vector regression to 0.11 and the neural network to 0.29, and even the linear baseline only reaches 0.37 (Fig. 6). The collapse is visible as a blank field in the map, since the two states are reconstructed so similarly that the difference carries no structure.

Predicting anomalies instead of absolute values removes most of this contrast. When each predictor field is expressed as a departure from the pre-industrial climatology of the target model, every learner recovers the mid-Holocene change at a correlation between 0.85 and 0.92, kernel ridge and the support vector regression included. The interannual correlations converge in the same way, between 0.49 and 0.55 for all seven methods, which is below the raw value of the tree methods and well above the raw value of the others. Transferability is thus governed mainly by whether the inter-model difference in base state is removed before the learning is applied, and not by the learner family alone. The tree methods remain the most robust choice because they are the only ones that work in both framings, and gradient boosting is used for the main results for that reason.

2.4 Absolute fields and the amplitude error

The interannual and difference metrics both cancel any constant offset, so they say nothing about whether the emulator reproduces $\delta^{18}O$ in absolute terms. The pre-industrial field answers that question directly. On the training model the emulated field is indistinguishable from the truth, with a spatial correlation of 1.00 and root-mean-square errors between 0.3 ‰ and 0.8 ‰ across the seven methods, evaluated on unseen years only. On the independent MPI-ESM-wiso the spatial pattern survives intact while the amplitude does not (Fig. 7). Every method keeps a spatial correlation of at least 0.97, yet the error grows to 1.2 ‰ for the random forest, 1.6 ‰ for gradient boosting and 2.0 ‰ for histogram gradient boosting, and to between 4.0 ‰ and 6.4 ‰ for the support vector regression, the neural network, kernel ridge and the linear baseline. The zonal profiles show where the error sits, in a systematic enrichment of the non-tree reconstructions through the tropics and the northern mid-latitudes.

This is a different failure mode from the one seen in the interannual metric. Shape is easy and level is hard, so an emulated absolute field should be read with a stated uncertainty of one to two per mil even where the pattern correlation looks perfect. The anomaly framing cannot be used here at all. Its output lives in the reference frame of the training model and retains a spatial standard deviation of only 0.43 ‰ against 14.33 ‰ in the truth, so absolute fields are available from the raw framing alone.

2.5 Mid-Holocene across the PMIP4 ensemble

Applying the emulator to nine PMIP4 models that carry no isotopes yields a mid-Holocene minus pre-industrial $\delta^{18}O$ field for each of them. The mean over the nine models and the seven methods is dominated by a depletion over northern Africa and the Sahel that exceeds 2 ‰ at its core, with a second centre over the Tibetan Plateau and central Asia (Fig. S4). At least seven of the nine models agree on the sign of the change over 43 % of the land surface, including the whole Saharan and Sahelian centre. The African depletion is the isotopic expression of the strengthened West African monsoon and the wetter Sahara of the mid-Holocene (Tierney et al., 2017; Brierley et al., 2020), and weak enrichment appears over South America, Australia and parts of the southern high latitudes.

The ensemble also separates the two sources of uncertainty in an emulated product. Changing the climate model shifts the reconstructed change by 0.20 ‰ in the land mean, while changing the learning method shifts it by 0.07 ‰ (Fig. S4). Drawn on a shared colour scale the two panels make the ratio readable as area, and the choice of model matters about three times more than the choice of learner. Both spreads peak over the same regions, the Sahara, the Tibetan Plateau and the East African highlands, where the mid-Holocene signal is largest and the climate-to-$\delta^{18}O$ mapping is least constrained. The raw framing gives a larger across-method spread of 0.28 ‰, but that number is inflated by the kernel methods that lose the change signal entirely, so the anomaly framing is the honest basis for the comparison.

2.6 Isotope–climate relations in the emulated fields

A final test asks what the emulated fields contain physically. Classical relations between $\delta^{18}O$ and climate are diagnosed in the same way from the two models that simulate isotopes explicitly and from the nine emulated PMIP4 models, in six latitude bands, both as a spatial gradient across grid points and as a temporal slope between the mid-Holocene and the pre-industrial state (Fig. 9). Agreement between the two groups indicates that the relation belongs to the simulated climate rather than to a prior the emulator carried over from its training model.

The spatial temperature effect passes this test. The emulated models reproduce the observed latitude dependence without being told about it, with slopes near 0.6 ‰ °C^{-1} at high northern latitudes, between 0.55 and 0.70 ‰ °C^{-1} at high southern latitudes, and a collapse to about 0.1 ‰ °C^{-1} in the tropics, all within the range spanned by AWIESM2-wiso and MPI-ESM-wiso. The spatial amount effect behaves the same way, near −1.1 ‰ per unit of log precipitation in the northern mid-latitudes for the emulated models against −1.06 ‰ in the truth.

The temporal slopes tell a more cautionary story. They run systematically below the spatial ones, with a median ratio of 0.49 over the 66 combinations of source and band, and the shortfall is far from uniform. The ratio is 0.65 between 60 and 90° N and 0.47 between 60 and 90° S, but only 0.31 between 30 and 60° N and 0.11 between 30 and 60° S. A palaeothermometer calibrated on the modern spatial gradient would therefore understate a mid-Holocene temperature change by a factor of two at high latitudes and by much more in the mid-latitudes. The emulator also reproduces the temporal slopes less faithfully than the spatial ones. The scatter of the emulated models around the truth is wider in the temporal direction, and between 0 and 30° S the emulated temperature slope even changes sign relative to the truth. The emulated fields are consequently reliable for present-day gradients and large-scale change patterns, and should not be used to calibrate a band-by-band temperature reconstruction.

3 Discussion

The emulator transfers between isotope-enabled models with a skill that has not been demonstrated before. Wider et al. (2023) found that a neural network trained on one model lost most of its skill when applied to another, falling to the level of pixel-wise linear regression. The emulator developed here keeps a land-mean temporal correlation of 0.66 and reproduces the mid-Holocene change at a spatial correlation of 0.87 on a model it never saw. The results here reproduce that earlier failure and also explain it, since the neural network and the two kernel methods behave in exactly the reported way while the tree ensembles do not. Two differences separate the two outcomes. The emulator learns from physical fields at single grid points instead of gridded images, so it never acquires the model-specific spatial structure that a convolutional network must learn. Geographic coordinates are also excluded from the predictors, and dropping them raises the cross-model temporal correlation slightly, from 0.63 to 0.65 for the mid-Holocene, at the cost of lowering the change correlation from 0.91 to 0.84. The emulator can therefore stand in for an explicit isotope scheme when a model does not carry one, provided the learner is chosen with transfer in mind.

The more general lesson concerns how skill is measured. A learner that fits the training model tightly can be worthless on a foreign one, and the ranking of the seven methods by internal validation error is close to the reverse of their ranking by cross-model skill. Any emulator intended for a model other than the one it was trained on should therefore be selected on an independent model rather than on a held-out split of its own training data. The same caution applies to hyperparameter tuning, which improved the internal score of two methods here without improving their transfer.

Removing the inter-model base state matters more than the choice of learner. In the raw framing the kernel methods lose the mid-Holocene signal completely, with correlations near zero, while the tree ensembles reach 0.84 to 0.87. Expressing the predictors as departures from the pre-industrial climatology of the target model lifts every method to between 0.85 and 0.92, kernel ridge included. What distinguishes the tree methods is thus not that they alone transfer, but that they are the only family that works in both framings, which makes them the safe default when a suitable baseline is unavailable. The anomaly framing has its own cost, since its output carries no absolute level and cannot deliver an absolute $\delta^{18}O$ field.

The attribution of uncertainty is correspondingly clearer than a raw comparison would suggest. Across the nine PMIP4 models the spread introduced by the choice of climate model is about three times that introduced by the choice of learning method. An earlier comparison based on the raw framing gave a much larger method spread, but that number is dominated by the two kernel methods that collapse under this framing and it overstates the real ambiguity of the product. Reporting an ensemble of learners remains worthwhile, since it exposes such a collapse where a single method would hide it, and the emulated change should still be read as a model-driven quantity rather than a method-driven one.

The absolute and the relative use of the emulator fail in different ways. Every method reproduces the spatial pattern of the pre-industrial field at a correlation of at least 0.97, yet the root-mean-square error on an independent model ranges from 1.2 ‰ for the random forest to 6.4 ‰ for the linear baseline. An emulated absolute field should consequently carry a stated uncertainty of one to two per mil even where its pattern looks perfect, while differences between two climate states are the more robust product. This distinction matters for proxy comparison, where absolute $\delta^{18}O$ and the change between periods are used for different purposes.

The isotope-climate relations diagnosed from the emulated fields support a physical reading of what the emulator learned, and at the same time set a limit on its use. The spatial temperature effect and its latitude dependence emerge in the nine emulated models without being prescribed, and they agree with the two models that simulate isotopes explicitly. The temporal slopes between the mid-Holocene and the pre-industrial state fall well below the spatial ones, by a median factor of two and by considerably more in the mid-latitudes. A palaeothermometer calibrated on modern spatial gradients would therefore understate past temperature change, a discrepancy already documented for ice cores (Jouzel et al., 1997) and now visible across an ensemble of models. The emulator reproduces these temporal slopes less faithfully than the spatial ones, so the emulated fields are best used for spatial patterns and large-scale change and not for a band-by-band temperature calibration.

Several limitations frame the interpretation. The emulator is trained on a single model, so its errors inherit the structural biases of that model, and the gap of about 0.1 in temporal correlation between the self-reconstruction and the cross-model test is the visible cost of this choice. Skill is lowest in arid regions, where precipitation is scarce and the precipitation-weighted $\delta^{18}O$ is noisy. The amount effect diagnosed in the polar bands is not a true amount effect, since polar precipitation and temperature are strongly collinear there and the regression attributes part of the temperature control to precipitation. The anomaly path requires the baseline of the target period to be taken from the target model itself, and a mismatched baseline collapses the mid-Holocene change signal. The comparison against real proxy archives, such as the speleothem records compiled in SISAL (Comas-Bru et al., 2020), is the natural next test and is left for future work. Isotope-enabled models have been compared with these records for the mid-Holocene and other periods (Parker et al., 2021; Bühler et al., 2021), and an emulator would extend the same comparison to the models that lack isotopes.

The approach extends naturally beyond the mid-Holocene. Training on additional isotope-enabled simulations would reduce the single-model dependence, and applying the emulator to colder periods such as the Last Glacial Maximum would test its behaviour outside the range of the training climate. A learned emulator that supplies consistent $\delta^{18}O$ for the full PMIP model ensemble would allow the many isotope-free models to be brought into direct comparison with the proxy record.

4 Methods

Training data. The emulator is trained on the isotope-enabled transient simulation eh2pi2.5, carried out with AWIESM2-wiso. The atmosphere component is ECHAM6-wiso, built on the general circulation model ECHAM6 (Stevens et al., 2013) and enhanced by the three stable water isotope tracers $H_2{}^{16}O$, $H_2{}^{18}O$ and HDO (Cauquoin and Werner, 2021; Shi et al., 2023). The ice-ocean component is FESOM2 (Danilov et al., 2017; Sidorenko et al., 2019). The atmospheric grid is T63L47, about 1.9° horizontal resolution with 47 vertical levels. The transient spans 8199 model years, with the first year at 8.2 ka BP and the last near 1950 CE. Monthly precipitation $\delta^{18}O$ is the prediction target. The predictors are fourteen climate fields, namely temperature, specific humidity and geopotential height at 850 and 500 hPa, the zonal wind at both levels and the meridional wind at 850 hPa, together with surface temperature, precipitation, vertically integrated cloud water, cloud cover and outgoing longwave radiation. Surface elevation and top-of-atmosphere insolation complete the predictor set. Latitude and longitude are deliberately excluded, so that the emulator cannot resolve a grid point by its position.

Validation and application data. Cross-model validation uses MPI-ESM-wiso (Cauquoin et al., 2019), which provides its own precipitation $\delta^{18}O$ for a mid-Holocene and a pre-industrial state. The mid-Holocene simulation contains an initial adjustment of roughly 150 years, so only its last 350 years are used, while the pre-industrial control is used in full at 353 years. The emulator is applied to nine PMIP4 mid-Holocene simulations, selected for having the required predictor fields, together with their pre-industrial controls (Otto-Bliesner et al., 2017; Brierley et al., 2020), each taken as a 100-year slice. All fields are regridded to the T63 grid of the training model.

Emulator and training. Seven regression methods from four learner families are trained through a common interface. The tree family comprises gradient boosting (Chen and Guestrin, 2016), a random forest (Breiman, 2001) and histogram gradient boosting. The kernel family comprises kernel ridge regression and support vector regression, both with radial basis functions and both fitted on a random subsample of at most fifteen thousand rows per region to keep the kernel computation tractable. A feed-forward neural network with two hidden layers of 128 and 64 units and a ridge regression complete the set. Separate models are fitted for each calendar month and for precipitation-based sub-regions of each continent, with the two ice sheets treated as single regions. One model year in eight enters the training set, which gives 1025 of the 8199 years, and the validation split is taken by whole years rather than by grid points so that no year contributes to both training and validation. Each training sample is weighted by its monthly precipitation, which aligns the loss with the precipitation-weighted product and down-weights the noisy dry-season samples. Where the surface lies above 2000 m the 850 hPa fields are treated as missing, since that level is below ground. The gradient-boosting hyperparameters are tuned with a time-block cross validation that maximises the held-out temporal correlation, and the tuned values are adopted only after they are confirmed to improve the independent cross-model skill.

Preprocessing. Two preprocessing paths are used. The raw path predicts absolute $\delta^{18}O$ from absolute predictor fields. The anomaly path expresses both the predictors and the target as departures from a climatology, taken during training as the mean of the last thousand years of the transient. When the emulator is applied to another model the baseline is recomputed from that model's own pre-industrial state, which aligns the two climates before the learned mapping is applied. A mismatched baseline removes the mid-Holocene signal, so this step is the critical one in the anomaly path. Insolation is retained in all configurations, because removing it flattens the orbital-scale trend of the transient.

Evaluation. Skill is evaluated on the precipitation-weighted annual $\delta^{18}O$ over land, as a per-grid-point temporal correlation of the annual series and as the spatial correlation and root-mean-square error of the multi-year climatology. Grid points with annual precipitation below 50 mm are excluded, since the precipitation-weighted isotope value is not meaningful where there is almost no precipitation, and the two ice sheets are exempt from this filter. Emulated values beyond 60 ‰ in magnitude are discarded as unphysical, which affects only the linear and neural methods at a small number of out-of-range grid points. Skill on the training model is computed after the 1025 training years are removed from the time axis, leaving 7174 unseen years. For the anomaly path both the emulated and the simulated series are referred to the same climatology before any comparison, since the anomaly output lives in the reference frame of the training model.

Isotope-climate relations. Slopes of $\delta^{18}O$ against surface temperature and against the logarithm of precipitation are computed in six latitude bands. The spatial slope is a regression across land grid points within a band for the pre-industrial state, and the temporal slope is the ratio of the mid-Holocene minus pre-industrial change in $\delta^{18}O$ to the corresponding change in temperature or log precipitation. The two isotope-enabled models and the nine emulated PMIP4 models are treated identically. AWIESM2-wiso has no separate mid-Holocene experiment, so a 1000-year window centred on 6 ka BP is taken from the transient and matched in length to the pre-industrial window formed by its last thousand years, with training years removed from both.

Data and code availability

The emulator code and the reconstruction outputs are available in the project repository. The training simulation and the MPI-ESM-wiso validation data are available from the respective model archives.

References

L. Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.

C. M. Brierley, A. Zhao, S. P. Harrison, et al. Large-scale features and evaluation of the PMIP4-CMIP6 midholocene simulations. *Climate of the Past*, 16:1847–1872, 2020. doi: 10.5194/cp-16-1847-2020.

J. C. Bühler, C. Roesch, M. Kirschner, et al. Comparison of the oxygen isotope signatures in speleothem records and iHadCM3 model simulations for the last millennium. *Climate of the Past*, 17:985–1004, 2021. doi: 10.5194/cp-17-985-2021.

A. Cauquoin and M. Werner. High-resolution nudged isotope modeling with ECHAM6-wiso: Impacts of updated model physics and ERA5 reanalysis data. *Journal of Advances in Modeling Earth Systems*, 13:e2021MS002532, 2021. doi: 10.1029/2021MS002532.

A. Cauquoin, M. Werner, and G. Lohmann. Water isotopes – climate relationships for the mid-Holocene and preindustrial period simulated with an isotope-enabled version of MPI-ESM. *Climate of the Past*, 15:1913–1937, 2019. doi: 10.5194/cp-15-1913-2019.

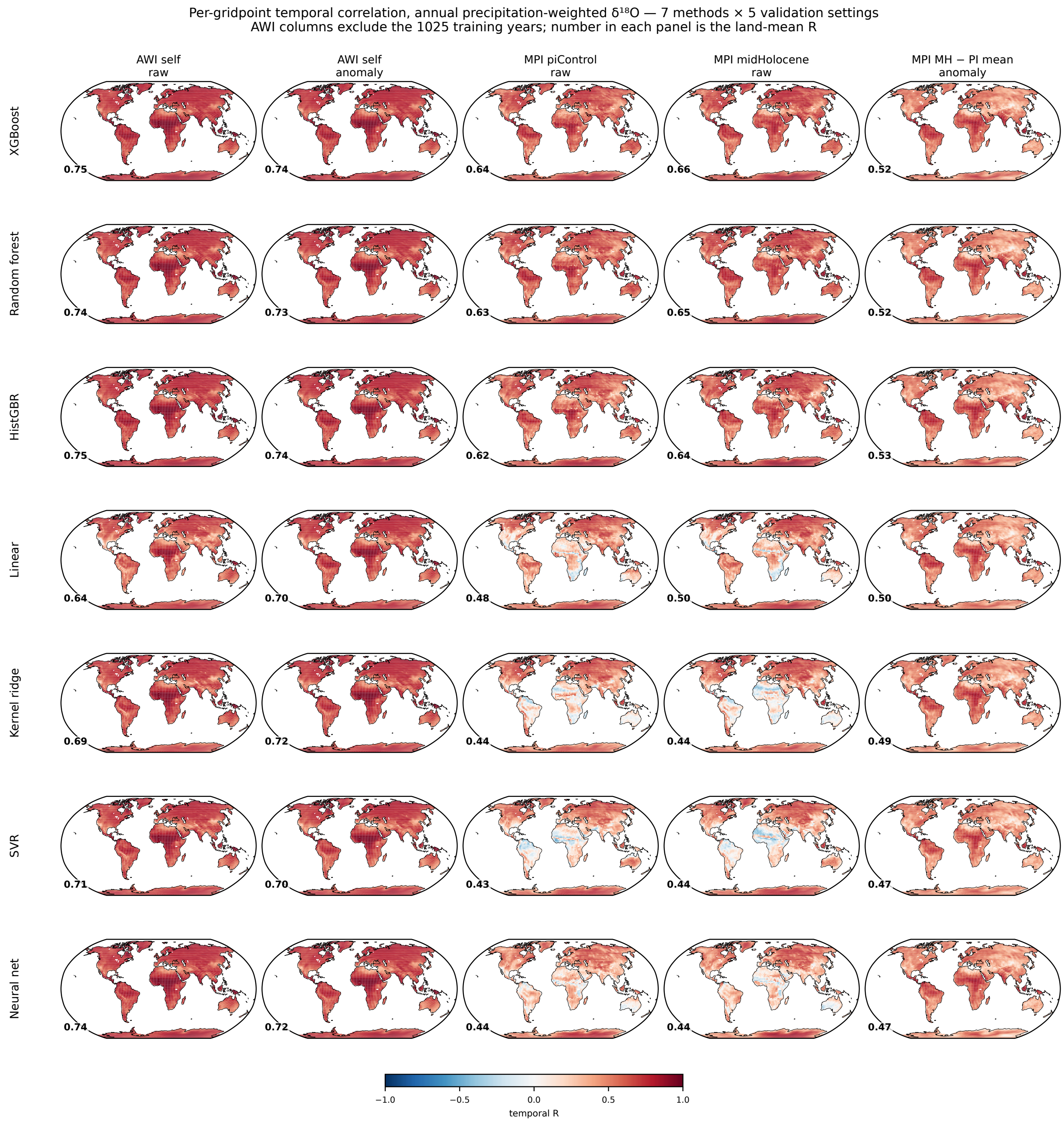

Figure 1: Per-grid-point temporal correlation between the emulated and the simulated annual precipitation-weighted $\delta^{18}O$ of the AWIESM2-wiso transient, for (a) gradient boosting, (b) random forest, (c) histogram gradient boosting, (d) ridge regression, (e) kernel ridge, (f) support vector regression and (g) a feed-forward neural network. Training years are excluded, leaving 7174 years. Land-mean correlation given in each panel. (h) Zonal mean of the correlation for all seven methods.

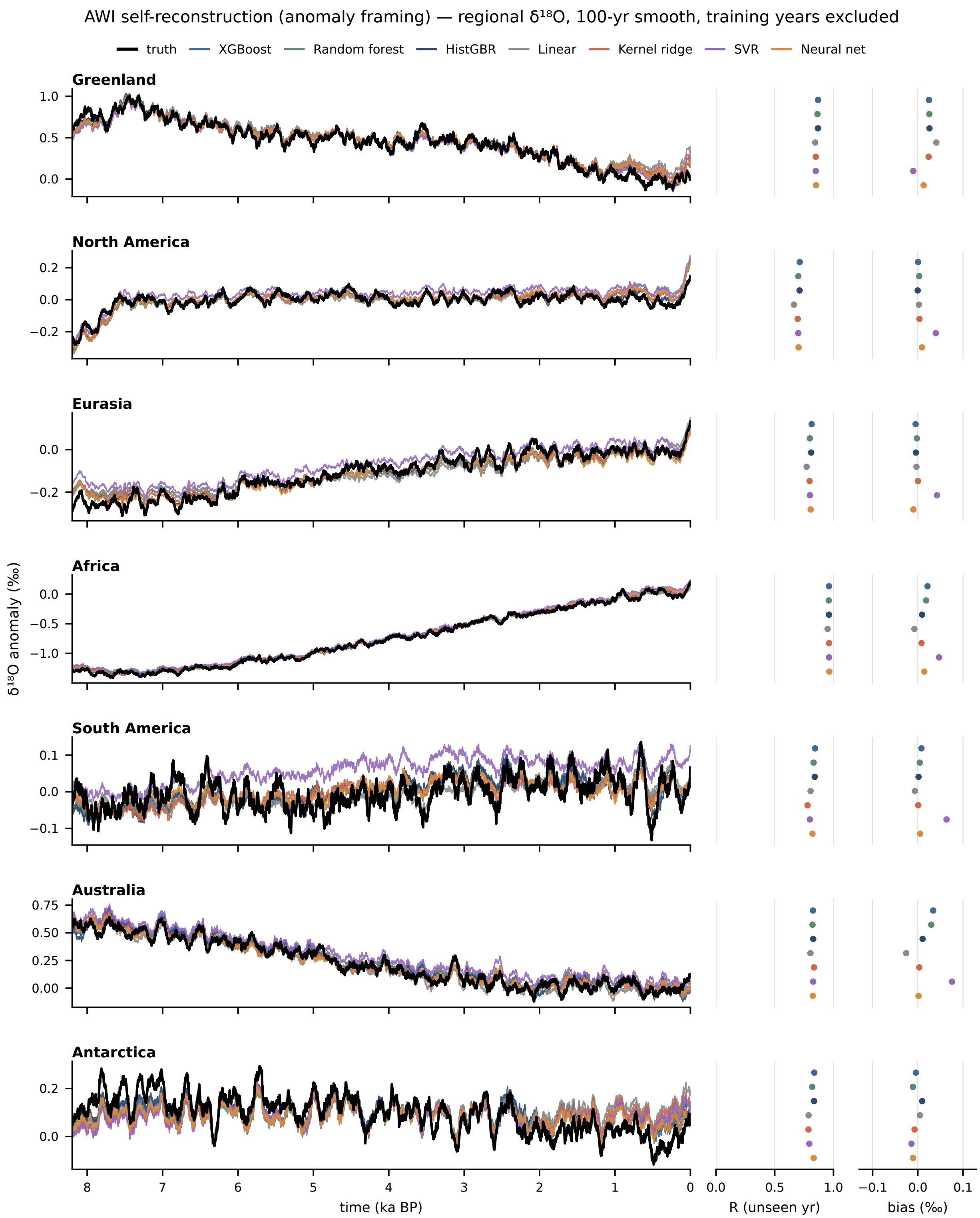

Figure 2: PLACEHOLDER CAPTION – regional time series, AWI self-reconstruction, anomaly framing. Seven regions north to south, truth and all seven learners, with per-region correlation and bias strips on the right. Training years are excluded.

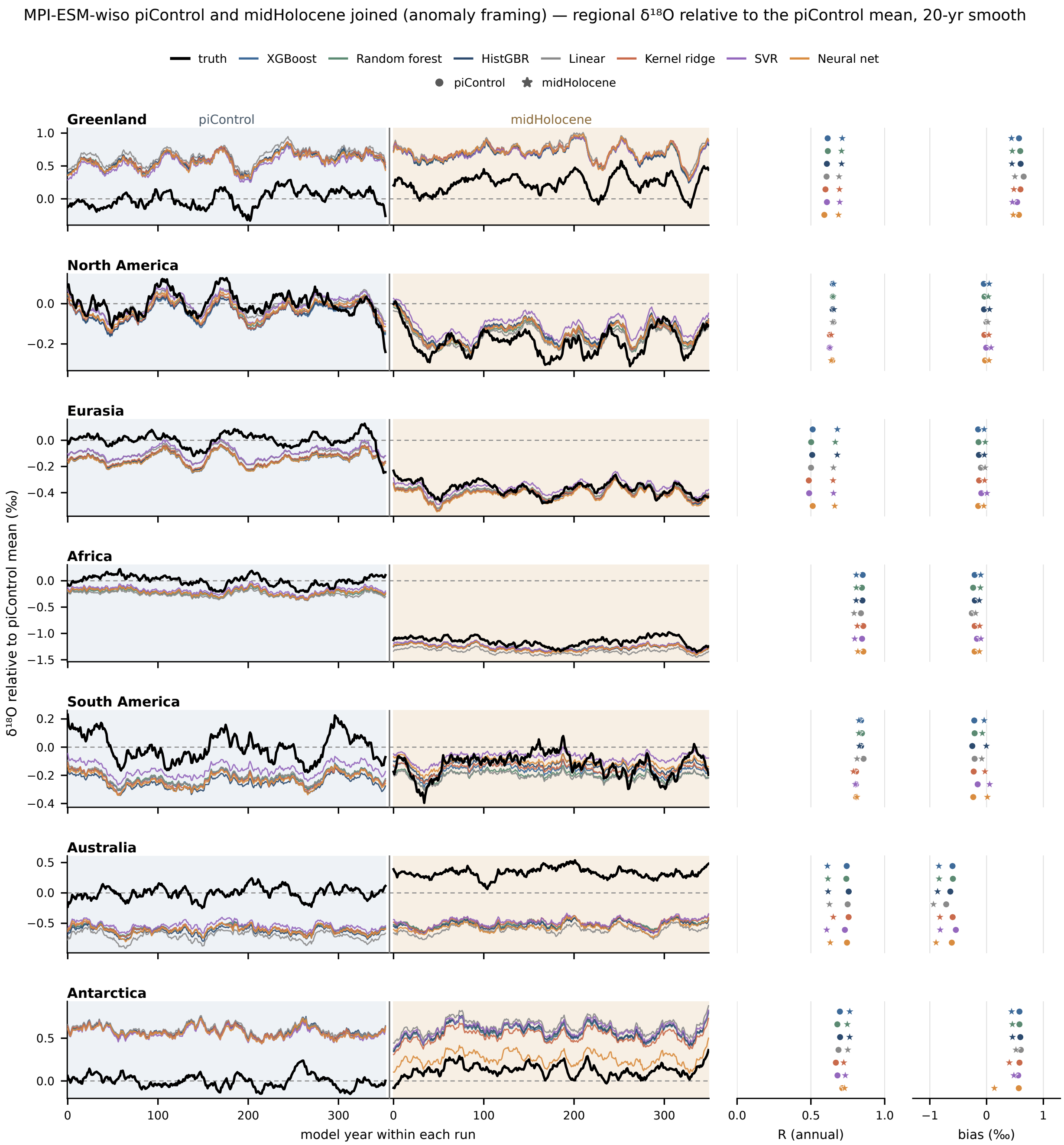

Figure 3: PLACEHOLDER CAPTION – MPI-ESM-wiso piControl and mid-Holocene joined into one series per region, $\delta^{18}O$ relative to the piControl mean state.

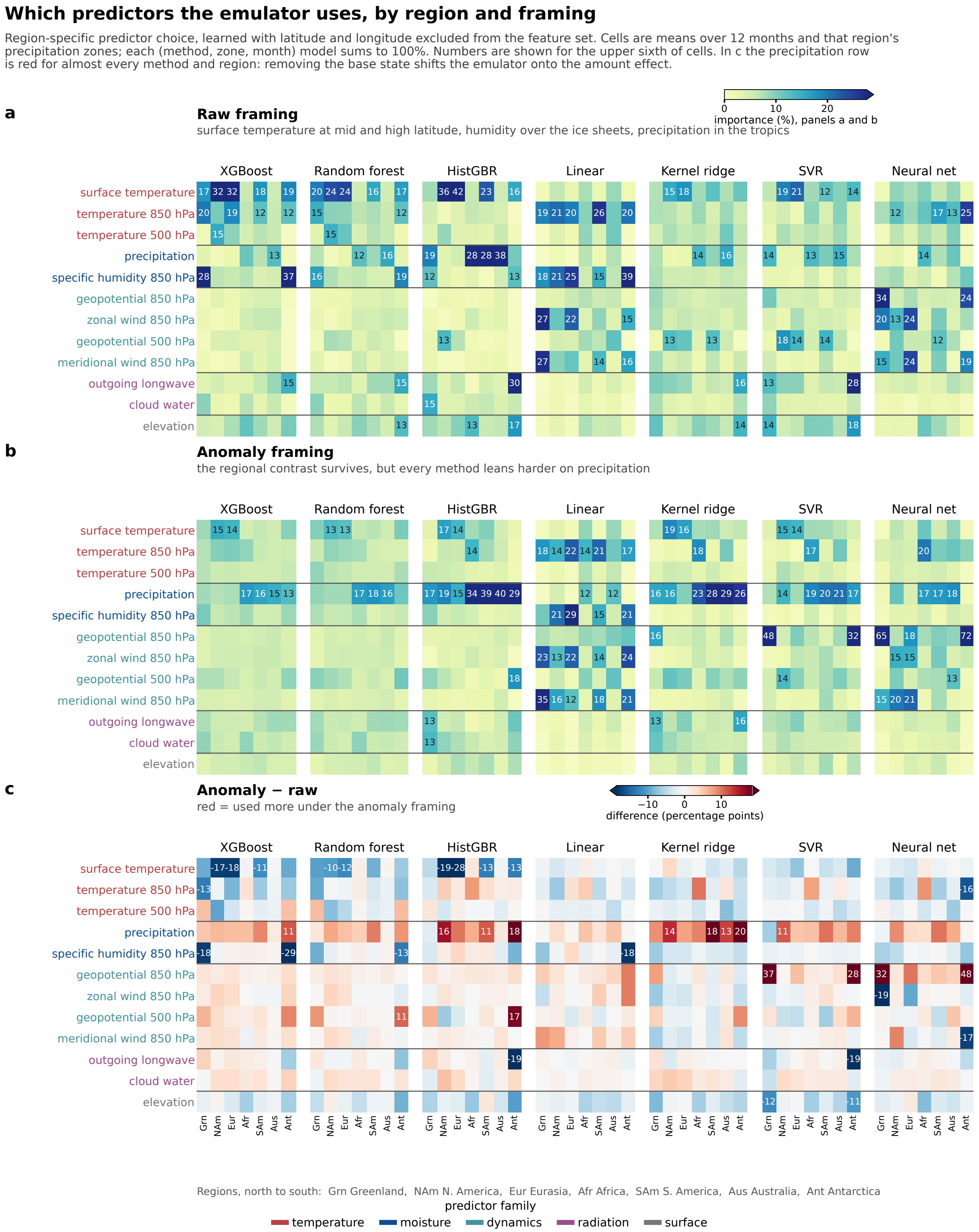

Figure 4: PLACEHOLDER CAPTION – predictor importance by method, region and framing, with latitude and longitude excluded from the feature set.

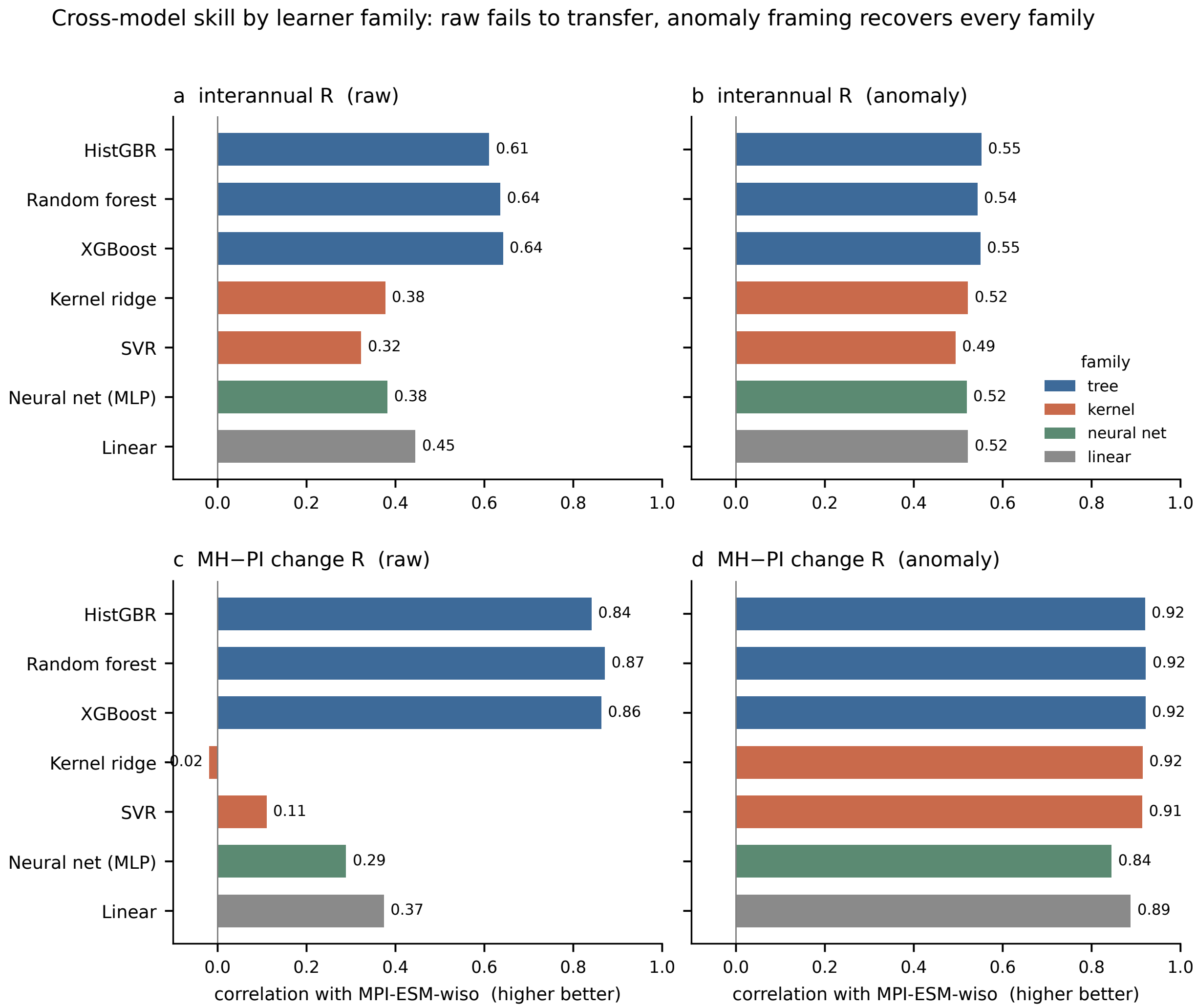

Figure 5: Cross-model skill of seven learners from four families, gradient boosting, random forest and histogram gradient boosting (tree), kernel ridge and support vector regression (kernel), a feed-forward neural network, and ridge regression (linear). (a, b) Precipitation-weighted temporal correlation on MPI-ESM-wiso for the raw and the anomaly framing. (c, d) As in (a)–(b), but for the spatial correlation of the mid-Holocene minus pre-industrial change.

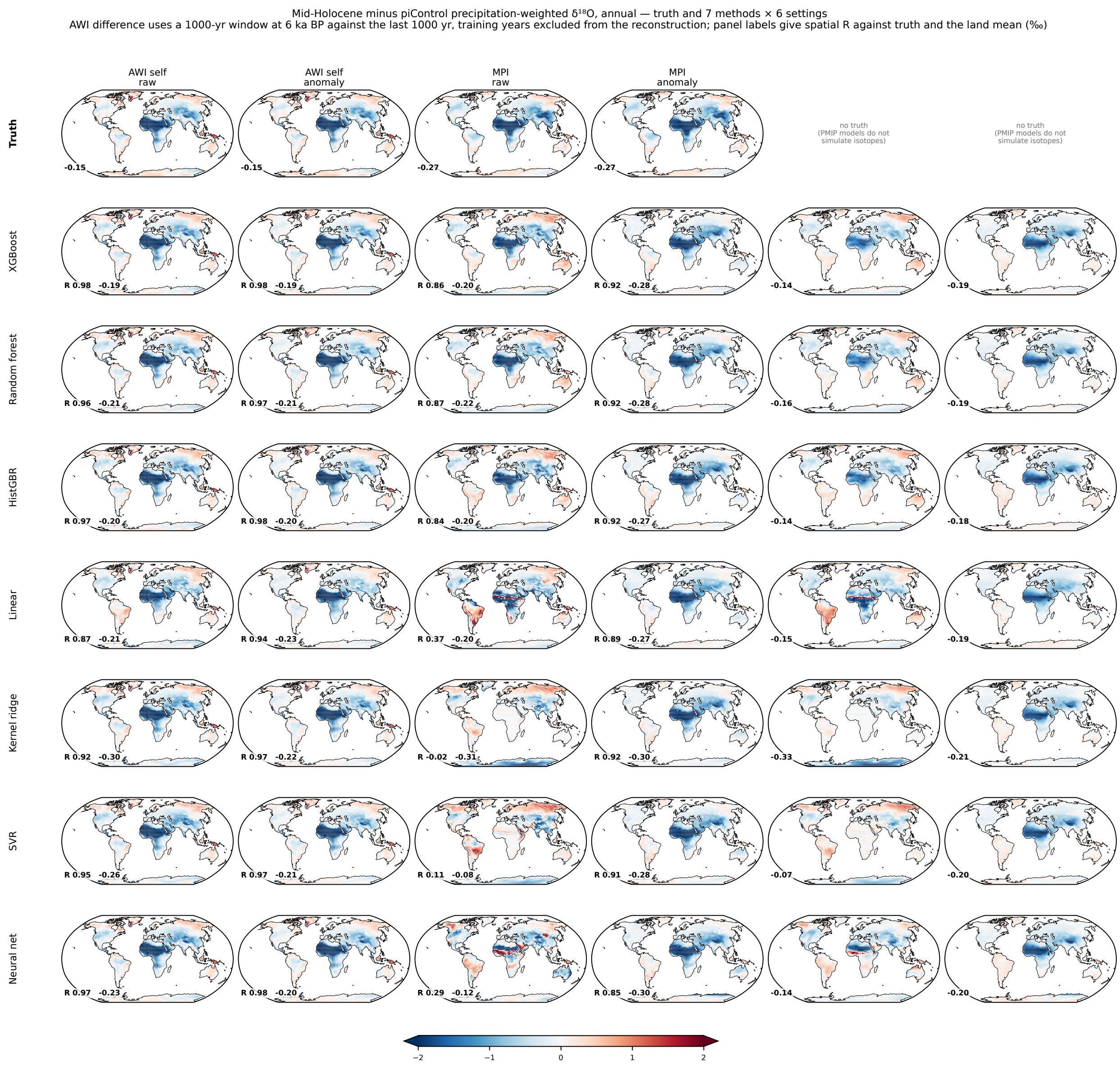

Figure 6: Mid-Holocene minus pre-industrial precipitation δ^{18}O in MPI-ESM-wiso, raw framing. (a) Simulated change. (b)–(h) Emulated change for the seven learners, with the spatial correlation against (a) given in each panel. (i) Zonal mean of the change. Units: ‰.

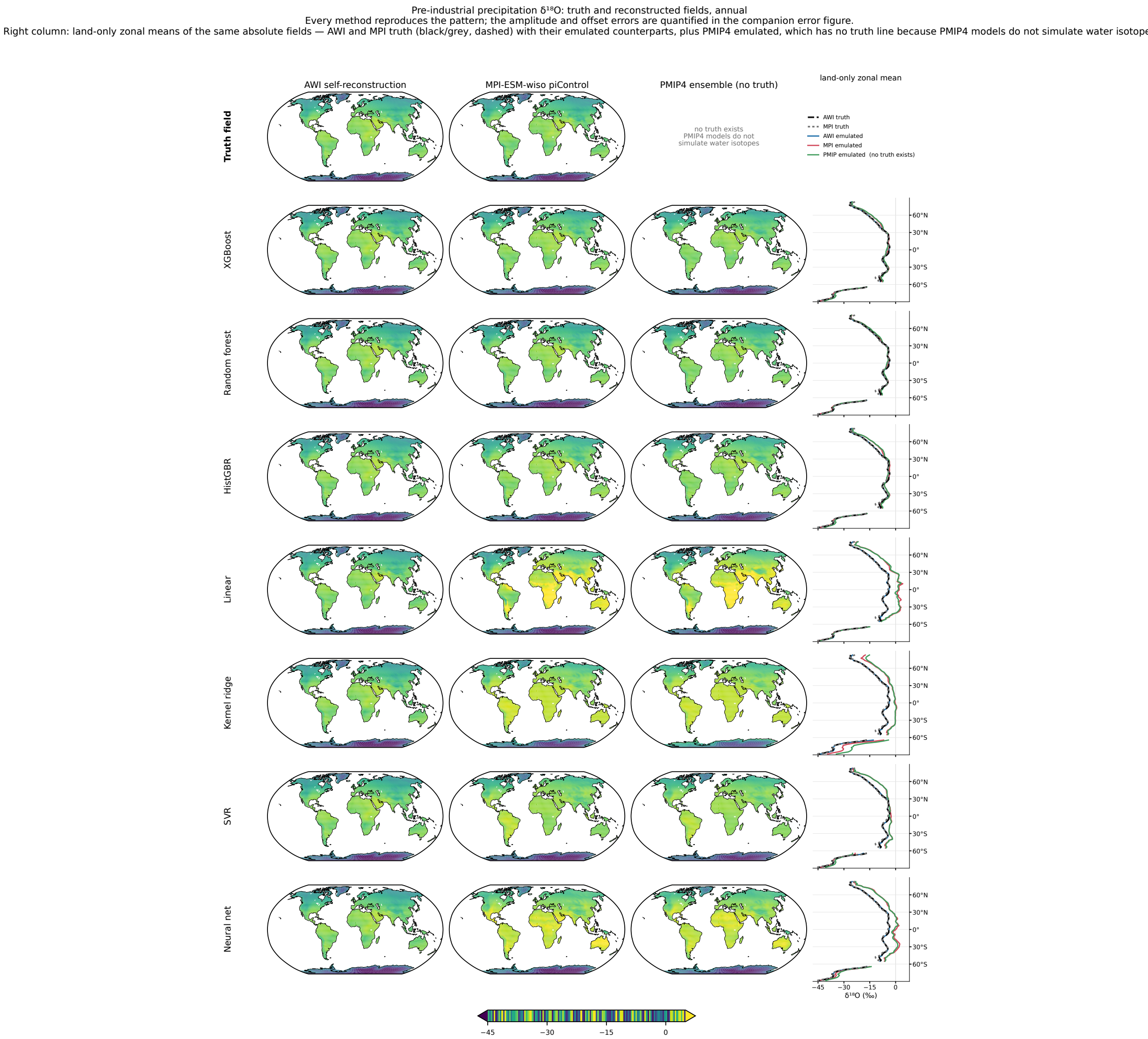

Figure 7: Pre-industrial precipitation δ^{18}O of MPI-ESM-wiso. (a) Simulated field. (b)–(h) Emulated field for the seven learners, with the spatial correlation and the root-mean-square error against (a) given in each panel. (i) Zonal mean. Units: ‰.

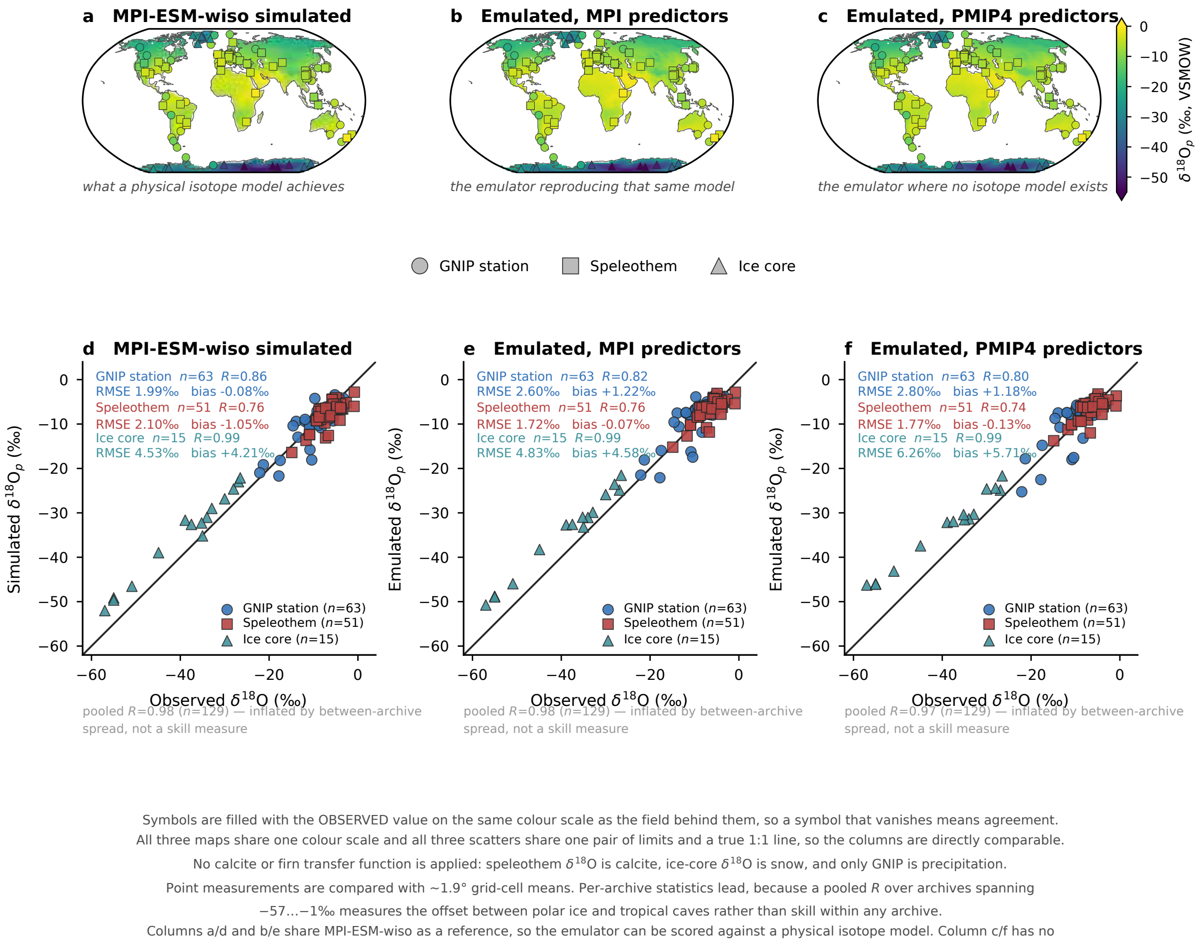

Figure 8: PLACEHOLDER CAPTION – Pre-industrial precipitation $\delta^{18}O$ against 129 proxy sites from three archives, 63 GNIP stations, 51 speleothem entities and 15 ice cores. (a, d) Isotope-enabled MPI-ESM-wiso, the reference a physical model achieves. (b, e) The emulator driven by the MPI-ESM-wiso climate fields. (c, f) The emulator driven by the PMIP4 climate fields, where no isotope-enabled counterpart exists. Statistics are given per archive; the pooled correlation is inflated by the offset between polar ice and tropical caves and is not a skill measure. Speleothem $\delta^{18}O$ is calcite reported on the VPDB scale and ice-core $\delta^{18}O$ is snow, whereas the emulator predicts precipitation $\delta^{18}O$ on the VSMOW scale. No calcite or firn transfer function is applied and no scale conversion is made, since neither converts one material into the other, so the absolute offsets in this figure, in particular the speleothem and ice-core biases, carry a material and scale component that is not emulator error.

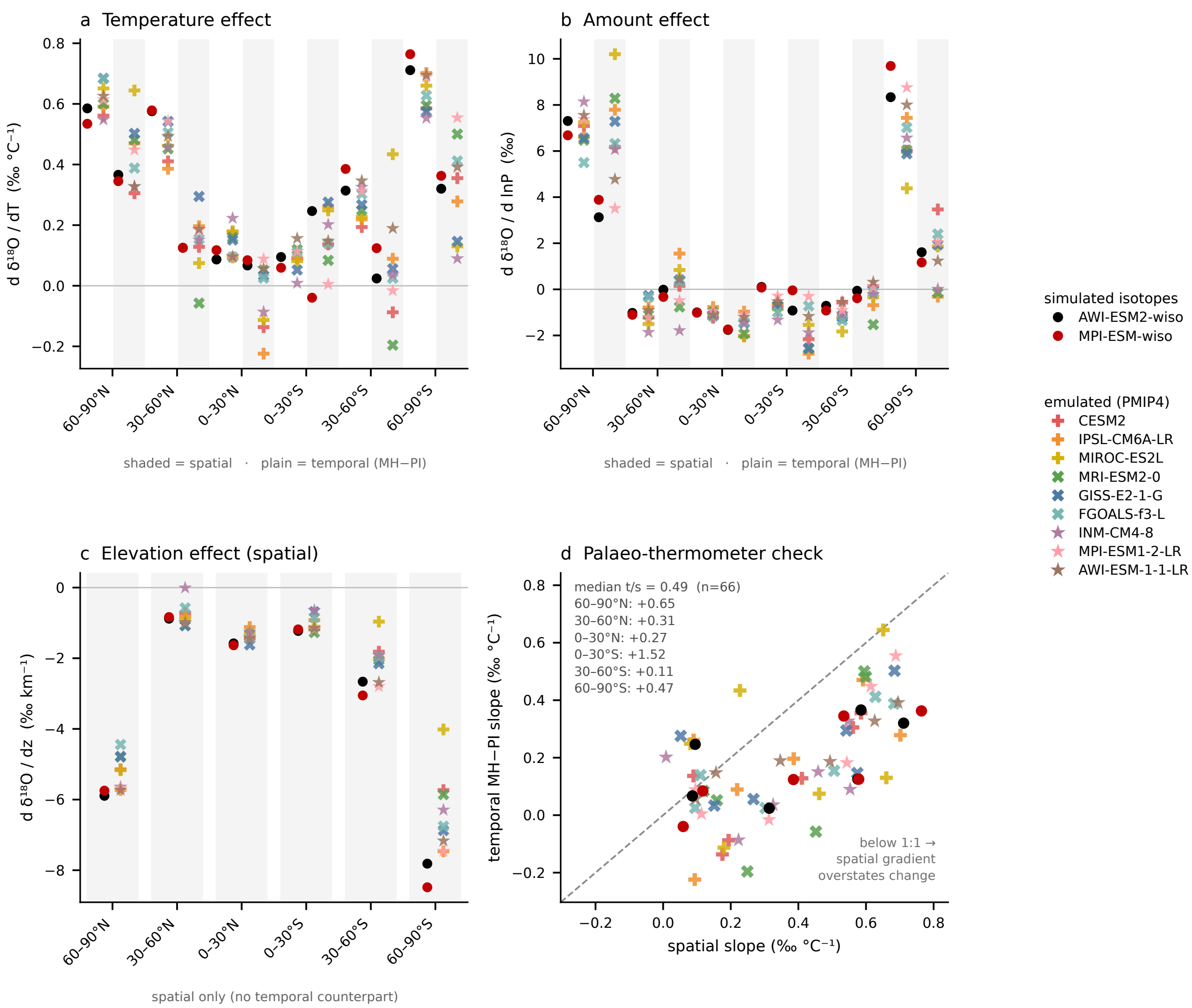

Figure 9: Isotope–climate relations by latitude band. (a) Slope of precipitation δ^{18}O against surface temperature, shaded columns for the spatial gradient across grid points and plain columns for the temporal mid-Holocene minus pre-industrial slope. (b) As in (a), but for the slope against the logarithm of precipitation. (c) Temporal against spatial temperature slope, one symbol per model and band, with the 1:1 line and the median ratio per band. Filled symbols mark the two models that simulate water isotopes explicitly and open symbols the nine emulated PMIP4 models.

T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

L. Comas-Bru, K. Rehfeld, C. Roesch, et al. SISALv2: a comprehensive speleothem isotope database with multiple age–depth models. *Earth System Science Data*, 12:2579–2606, 2020. doi: 10.5194/essd-12-2579-2020.

N. Cresswell-Clay, B. Liu, D. Durran, et al. A deep learning Earth system model for efficient simulation of the observed climate. *AGU Advances*, 6:e2025AV001706, 2025. doi: 10.1029/2025AV001706.

Y. Cui, L. Tian, Z. Cai, and S. Wang. Spatially inhomogeneous response of precipitation $\delta^{18}O$ in China to ENSO cycles. *npj Climate and Atmospheric Science*, 8:31, 2025. doi: 10.1038/s41612-025-01057-1.

S. Danilov, D. Sidorenko, Q. Wang, and T. Jung. The Finite-volumE Sea ice–Ocean Model (FESOM2). *Geoscientific Model Development*, 10:765–789, 2017. doi: 10.5194/gmd-10-765-2017.

W. Dansgaard. Stable isotopes in precipitation. *Tellus*, 16:436–468, 1964. doi: 10.3402/tellusa.v16i4.8993.

S. Dee, J. Emile-Geay, M. N. Evans, et al. PRYSM: An open-source framework for PRoxY System Modeling, with applications to oxygen-isotope systems. *Journal of Advances in Modeling Earth Systems*, 7:1220–1247, 2015. doi: 10.1002/2015MS000447.

D. Erdélyi, I. G. Hatvani, H. Jeon, et al. Predicting spatial distribution of stable isotopes in precipitation by classical geostatistical- and machine learning methods. *Journal of Hydrology*, 617:129129, 2023. doi: 10.1016/j.jhydrol.2023.129129.

G. Falster, G. Abramowitz, S. Hobeichi, et al. High resolution monthly precipitation isotope estimates across Australia from machine learning. *Hydrology and Earth System Sciences*, 30: 289–315, 2026. doi: 10.5194/hess-30-289-2026.

G. J. Hakim, J. Emile-Geay, E. J. Steig, et al. The last millennium climate reanalysis project: Framework and first results. *Journal of Geophysical Research: Atmospheres*, 121:6745–6764, 2016. doi: 10.1002/2016JD024751.

J. Jouzel, R. B. Alley, K. M. Cuffey, W. Dansgaard, P. Grootes, G. Hoffmann, S. J. Johnsen, R. D. Koster, D. Peel, C. A. Shuman, M. Stievenard, M. Stuiver, and J. White. Validity of the temperature reconstruction from water isotopes in ice cores. *Journal of Geophysical Research: Oceans*, 102(C12):26471–26487, 1997. doi: 10.1029/97JC01283.

E. S. Klein, J. E. Cherry, J. M. Welker, et al. Complexity of Arctic Ocean water isotope spatial and temporal patterns revealed with machine learning. *Elementa: Science of the Anthropocene*, 12:00127, 2024. doi: 10.1525/elementa.2023.00127.

J. Liu, J. Ding, J. Wang, J. Zou, Z. Zhang, Q. Bao, X. Wang, and X. Ge. Predicting the isotopic composition of precipitation in China via interpretable machine learning techniques. *Journal of Hydrology*, page 134190, 2025. doi: 10.1016/j.jhydrol.2025.134190.

D. B. Nelson, D. Basler, and A. Kahmen. Precipitation isotope time series predictions from machine learning applied in Europe. *Proceedings of the National Academy of Sciences*, 118(26): e2024107118, 2021. doi: 10.1073/pnas.2024107118.

B. L. Otto-Bliesner, P. Braconnot, S. P. Harrison, et al. The PMIP4 contribution to CMIP6 – Part 2: Two interglacials, scientific objective and experimental design for Holocene and Last Interglacial simulations. *Geoscientific Model Development*, 10:3979–4003, 2017. doi: 10.5194/gmd-10-3979-2017.

S. E. Parker, S. P. Harrison, et al. A data–model approach to interpreting speleothem oxygen isotope records from monsoon regions. *Climate of the Past*, 17:1119–1138, 2021. doi: 10.5194/cp-17-1119-2021.

X. Shi, M. Werner, A. Cauquoin, H. Yang, P. Gierz, and G. Lohmann. Simulated stable water isotopes during the mid-Holocene and pre-industrial periods using AWI-ESM-2.1-wiso. *Geoscientific Model Development*, 16:5153–5178, 2023. doi: 10.5194/gmd-16-5153-2023.

D. Sidorenko, H. F. Goessling, N. V. Koldunov, et al. Evaluation of FESOM2.0 coupled to ECHAM6.3: Preindustrial and HighResMIP simulations. *Journal of Advances in Modeling Earth Systems*, 11:3794–3815, 2019. doi: 10.1029/2019MS001696.

B. Stevens, M. Giorgetta, M. Esch, et al. Atmospheric component of the MPI-M Earth System Model: ECHAM6. *Journal of Advances in Modeling Earth Systems*, 5:146–172, 2013. doi: 10.1002/jame.20015.

C. Tebaldi et al. Emulators of climate model output. *Annual Review of Environment and Resources*, 50:709–737, 2025. doi: 10.1146/annurev-environ-012125-085838.

J. E. Tierney, F. S. R. Pausata, and P. B. deMenocal. Rainfall regimes of the Green Sahara. *Science Advances*, 3:e1601503, 2017. doi: 10.1126/sciadv.1601503.

S. Wang, Y. Xiao, Y. Qian, H. Li, C. Che, X. Zhu, and M. Zhang. Reconstructed precipitation isotopes in China during the past six decades based on machine learning. *Journal of Hydrology*, 661:134003, 2025. doi: 10.1016/j.jhydrol.2025.134003.

J. Wider, J. Kruse, N. Weitzel, J. C. Bühler, U. Köthe, and K. Rehfeld. Towards learned emulation of interannual water isotopologue variations in General Circulation Models. *Environmental Data Science*, 2:e35, 2023. doi: 10.1017/eds.2023.29.

Supplementary figures

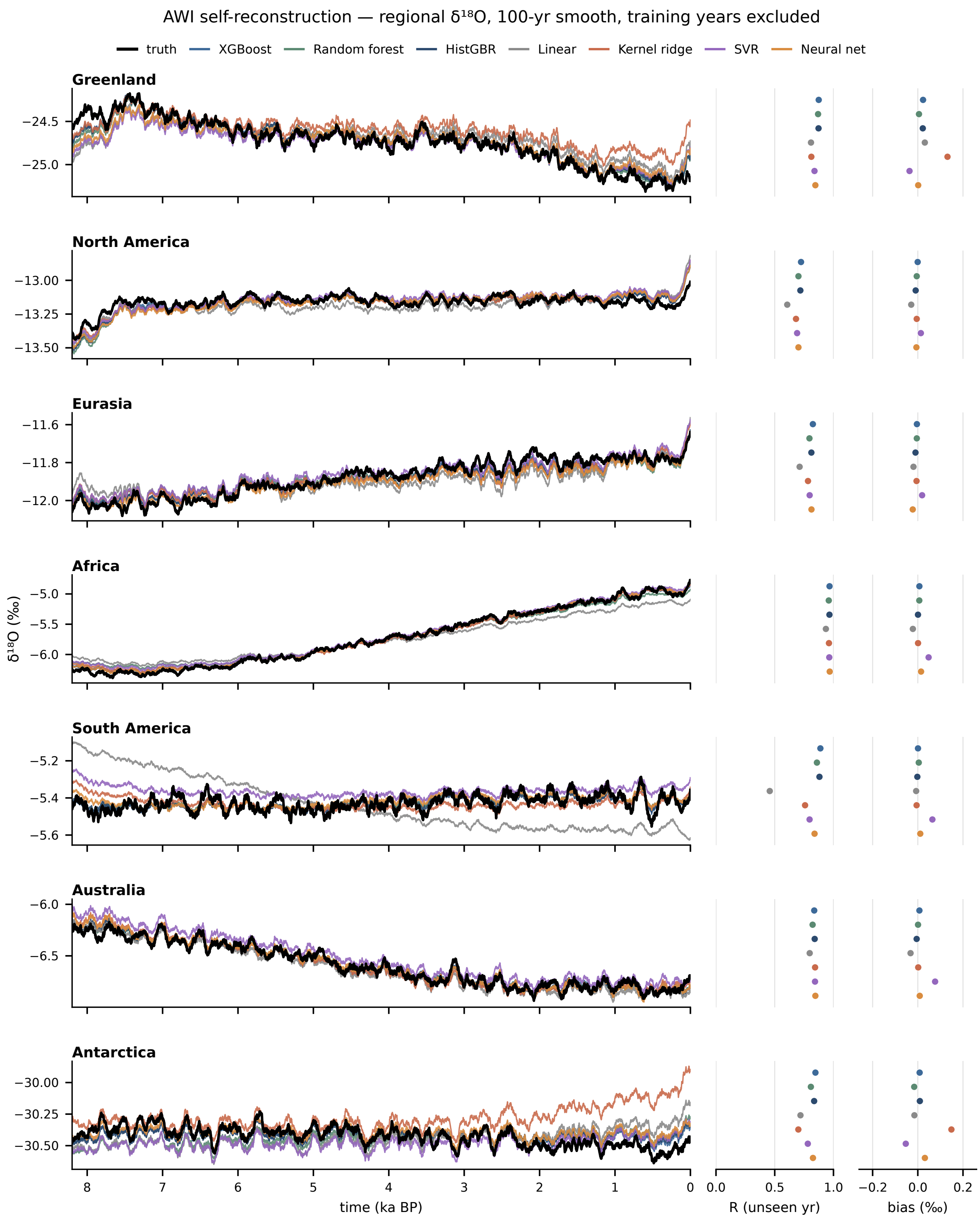

Figure S1: PLACEHOLDER CAPTION – AWI self-reconstruction, absolute regional $\delta^{18}O$ over the 8.2 kyr transient, seven regions north to south. Truth and all seven learners, with per-region correlation and bias strips. Training years are excluded from both the traces and the statistics.

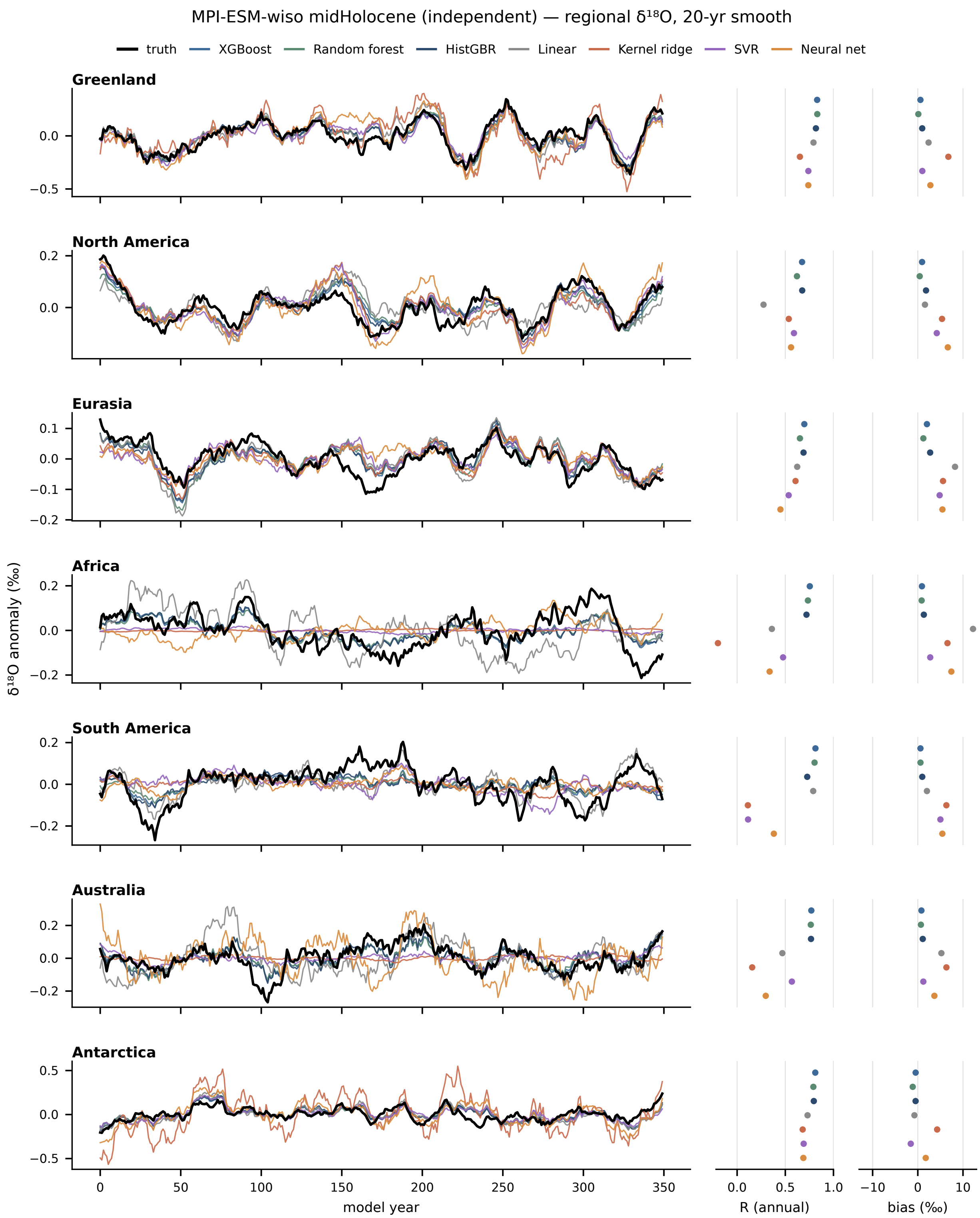

Figure S2: PLACEHOLDER CAPTION – MPI-ESM-wiso mid-Holocene, raw framing. Traces are de-meaned for display because the cross-model offset would otherwise flatten them; the offset itself is reported in the bias strip.

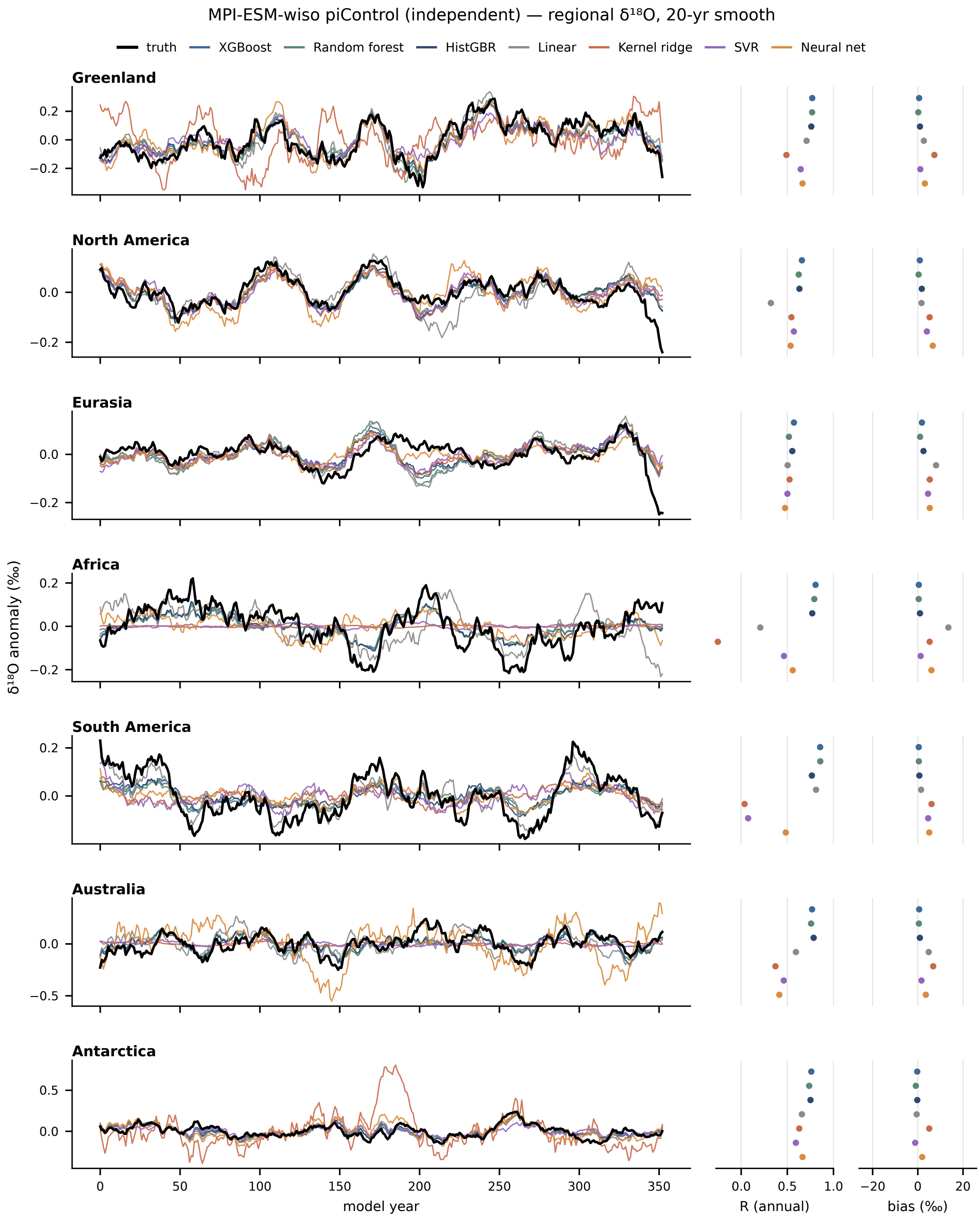

Figure S3: PLACEHOLDER CAPTION – As in Fig. S2, but for the piControl state.

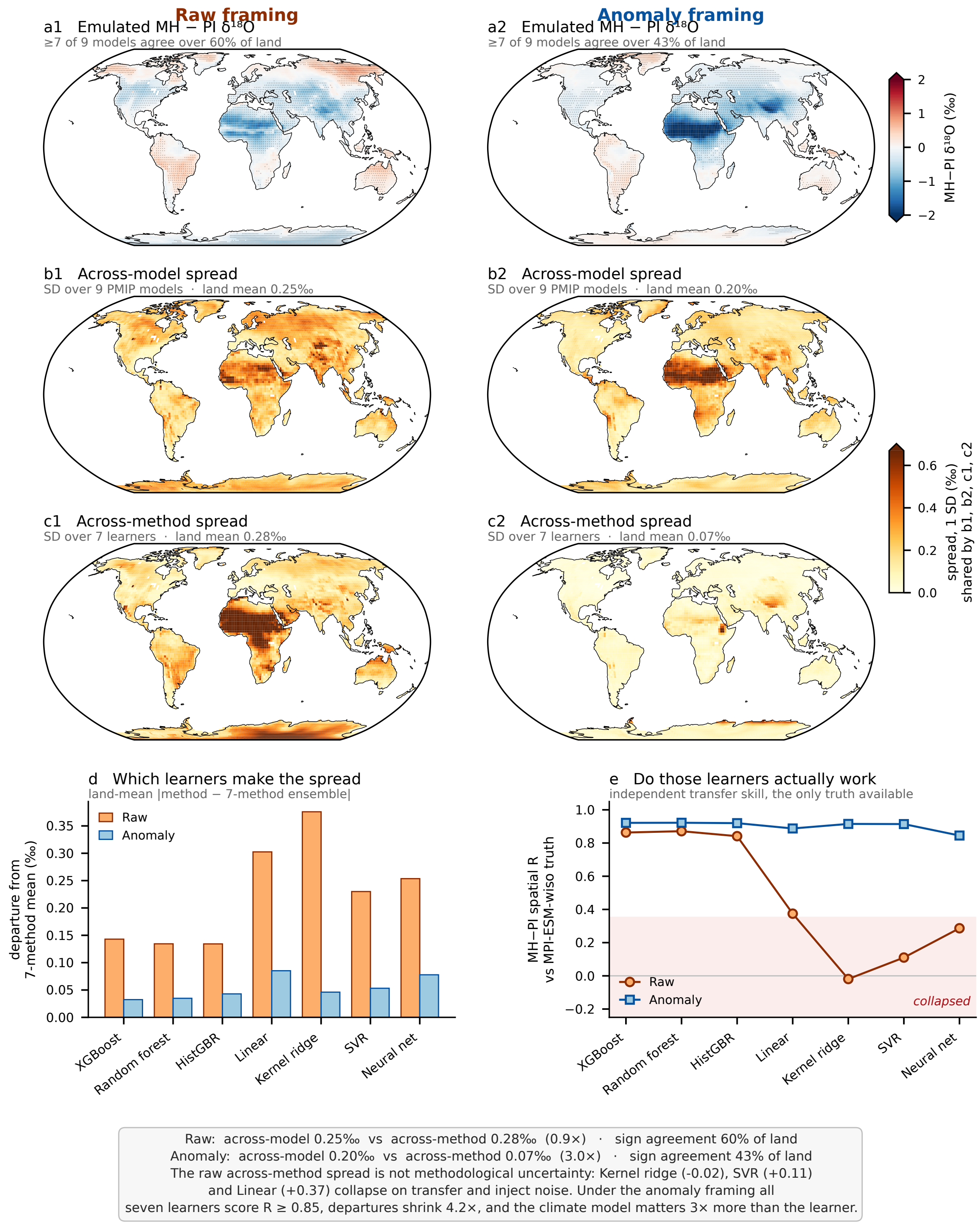

Figure S4: PLACEHOLDER CAPTION – Raw against anomaly framing for the PMIP4 ensemble. The larger across-method spread under the raw framing is produced by the learners that fail to transfer, not by genuine methodological uncertainty.

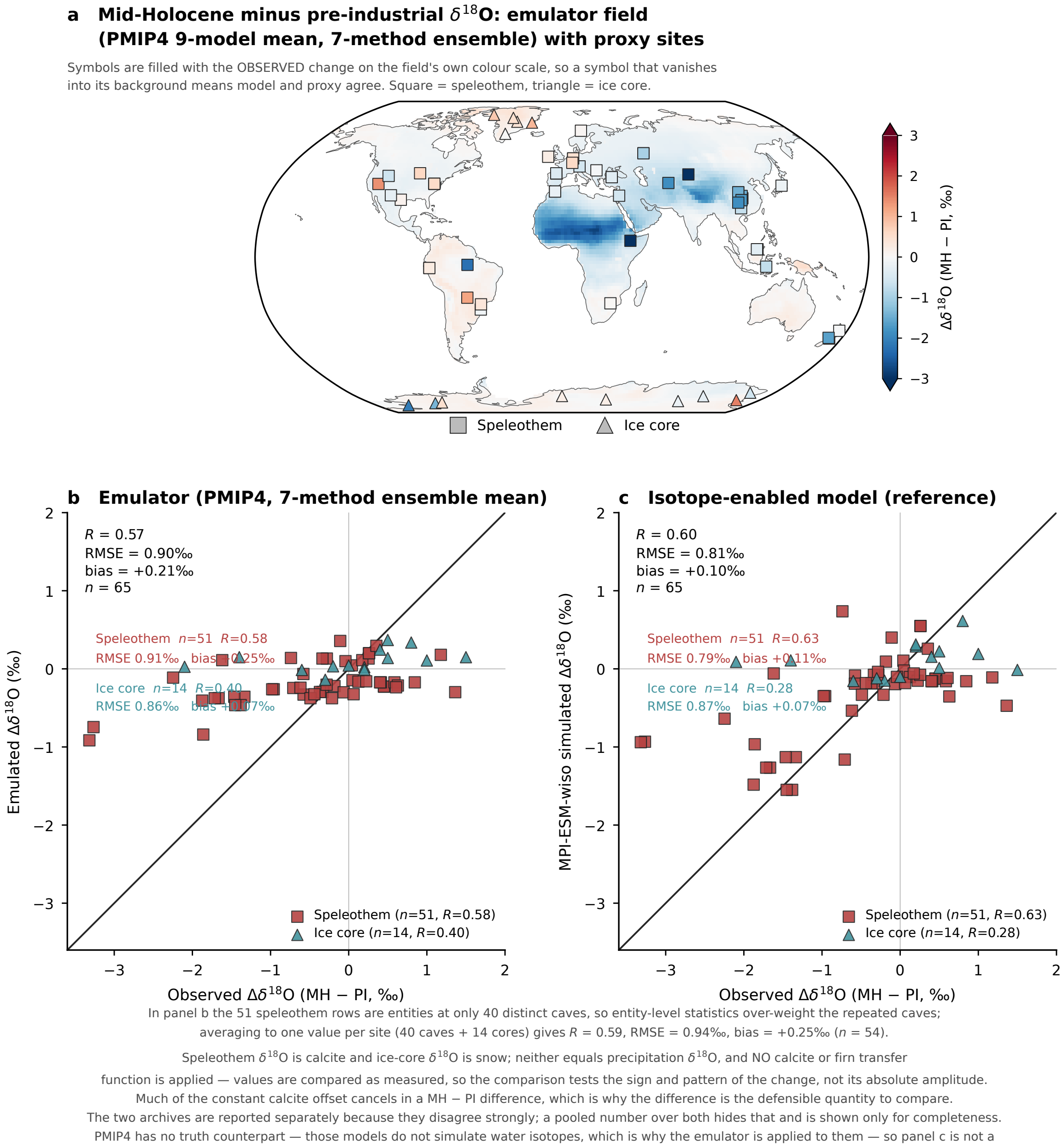

Figure S5: PLACEHOLDER CAPTION – Mid-Holocene minus pre-industrial δ^{18}O against 65 proxy sites, 51 speleothem entities and 14 ice cores, anomaly framing. (a) Emulated change, PMIP4 nine-model and seven-method ensemble mean, with the sites drawn on the colour scale of the field. (b) Observed against emulated change. (c) Observed against isotope-enabled MPI-ESM-wiso at the same sites, the reference a physical model achieves. Speleothem δ^{18}O is calcite reported on the VPDB scale and ice-core δ^{18}O is snow, whereas the emulator predicts precipitation δ^{18}O on the VSMOW scale. No calcite or firn transfer function is applied and no scale conversion is made. The comparison therefore tests the sign and the pattern of the change rather than its absolute amplitude, which is defensible here because a constant material offset cancels in the difference and a change of reference scale rescales it by about 3 %, that is by 0.01 ‰ on the observed mean change of −0.39 ‰, far below the error of 0.9 ‰.

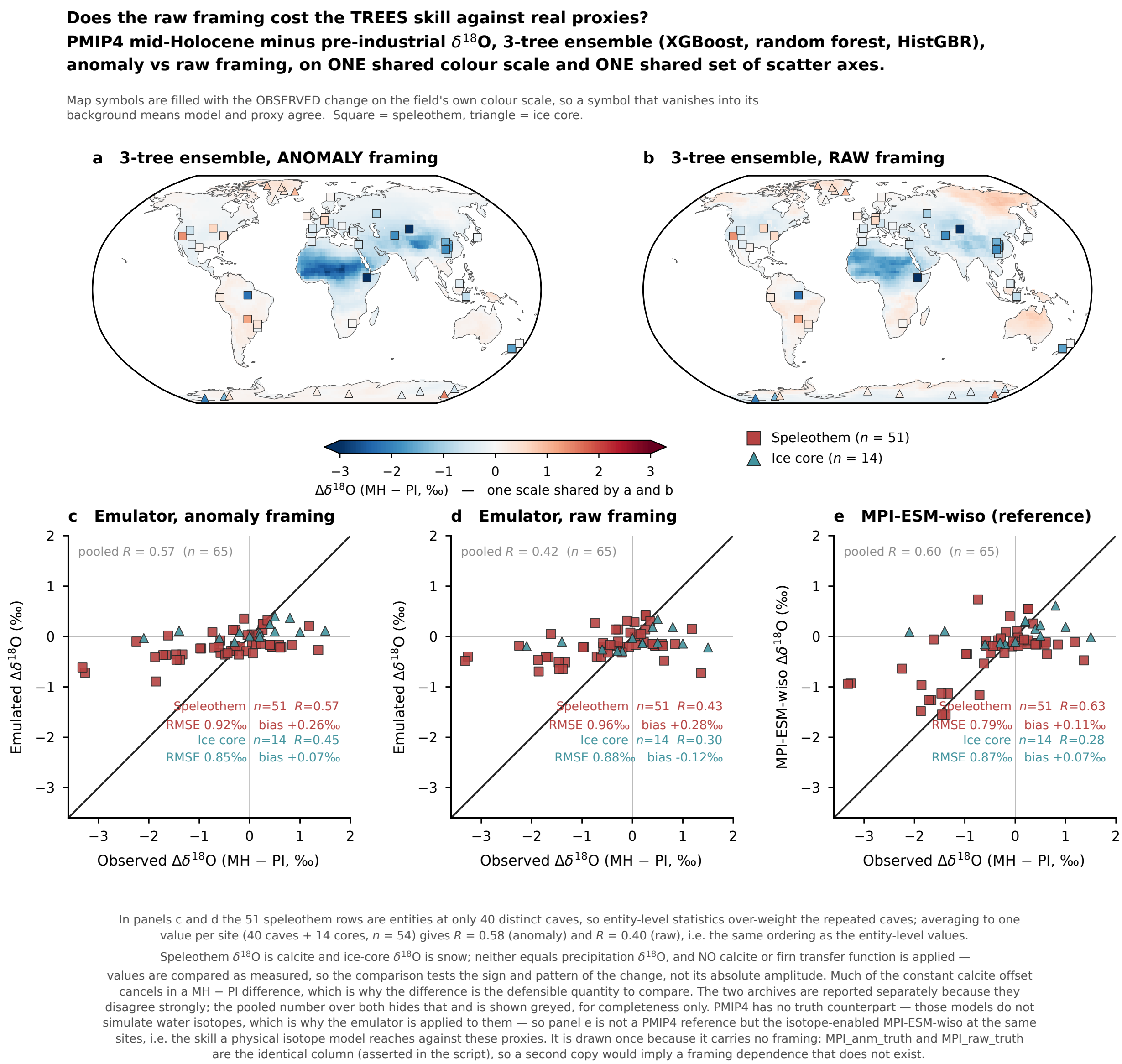

Figure S6: PLACEHOLDER CAPTION – The same 65 proxy sites used to test the framing rather than the learner family, with the ensemble held fixed at the three tree methods, which are the family that survives the raw framing. (a, c) Anomaly framing. (b, d) Raw framing, on the same colour scale and the same scatter axes. (e) Isotope-enabled MPI-ESM-wiso, drawn once because a physical isotope model carries no framing. Against real proxies the anomaly framing is better than the raw framing even for the tree methods, so the advantage of the anomaly framing established against models in Fig. 5 is not an artefact of the learners that fail to transfer. Speleothem $\delta^{18}O$ is calcite reported on the VPDB scale and ice-core $\delta^{18}O$ is snow, whereas the emulator predicts precipitation $\delta^{18}O$ on the VSMOW scale, and no calcite or firn transfer function and no scale conversion are applied, so the comparison tests the sign and the pattern of the change rather than its absolute amplitude.